

COMP3069 Computer Graphics

Coursework (Assessment 3) – 2024

Marking Scheme

Deliverable	Marks
Report with demo Marks are available for a correctly filled report, including: 1. All required screenshots are attached. (10) 2. All required descriptions are clear and accurate. (10) Please refer to the Report Sheet for the detailed marking scheme of the report part.	20
Tidy Code & directory structure Two marks are available for tidy code (1) & appropriate/clear comments (2).	3
General 3D graphics programming Marks are available for showing that you can use the appropriate APIs taught in COMP3069. This includes, but is not limited to: 1. GLFW code for window creation. (1) 2. GLFW code for handling input. (1) 3. C code for model-view-projection and fly-through-projection matrices. (1) 4. GLSL code for a vertex shader. (1) 5. GLSL code for a fragment shader. (1) 6. OpenGL code for VAO specification (position, normal, UV or colour). (1) 7. OpenGL code for Anti-aliasing. We will check to see if diagonal edges are jaggy. (1)	7
Task 1 - 3D modelling & 3D transformations You will receive marks for four objects that you model and render. You are expected to spend 40 hours of work on this requirement. The mark will be proportionate to the object complexity and similarity to those in Figure 1. Marks for each object include: 1. C code for vertex specification. (1) 2. OpenGL code for 3D Transformations (e.g., rotate, translate, scale). You need to apply transformations to achieve the desired effect. (2) 3. An appropriate appearance. This is to discourage you from modelling, e.g.,	16

a simple cube and claiming that it is a ball. (1)	
Task 2 - Lights You will receive 1 mark for each of 3 different (types of) lights. Marks include: 1. Specifying the light & using a uniform. (0.5) 2. The lighting calculations in the fragment shader. (0.5)	3
Task 3 - Textures You will receive marks for textures. Marks include: 1. Loading bitmap & create OpenGL texture. (1) 2. Specifying the texture coordinates. (1) 3. Magnification. (2) 4. Minification. (2)	6
Task 4 - Cameras Marks will be given for: 1. Model-Viewer Camera. (2) 2. Fly-Through Camera. (2)	4
Task 5 – Anti-aliasing Marks will be awarded for the implementation of anti-aliasing of edges.	1
More marks are given for others and the similarity to the scene (i.e., Figure 1), including: 1. The overall look and the connections of the objects are similar to those in Figure 1. (2) 2. Colour (texture) matching to all objects. (2) 3. Implementation of all possible interaction with the four movable/changeable objects. (2) 4. Implementation of the interaction with lights (e.g., switching on and off and between the lights). (2) 5. Implementation of animations of all four objects. (2)	10
Total Marks	70